

Review

Water Quality Management in the Age of AI: Applications, Challenges, and Prospects

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Abstract: Artificial intelligence (AI) is transforming water environment management, creating new opportunities for improved monitoring, prediction, and intelligent regulation of water quality. This review highlights the transformative impact of AI, particularly through hybrid modeling frameworks that integrate AI with technologies like the Internet of Things (IoT), Remote Sensing (RS), and Unmanned Monitoring Platforms (UMP). These advances have significantly enhanced real-time monitoring accuracy, expanded the scope of data acquisition, and enabled comprehensive analysis through multisource data fusion. Coupling AI models with process-based models (PBM) has notably enhanced predictive capabilities for simulating water quality dynamics. Additionally, AI facilitates dynamic early-warning systems, precise pollutant source tracking, and data-driven decision-making. However, significant challenges remain, including data quality and accessibility, model interpretability, monitoring of hard-to-measure pollutants, and the lack of system integration and standardization. To address these bottlenecks, future research should focus on: (1) constructing high-quality, standardized open-access datasets; (2) developing explainable AI (XAI) models; (3) strengthening integration with digital twins and next-generation sensors; (4) improving the monitoring of trace and emerging pollutants; and (5) coupling AI with PBM by optimizing input data, internal mechanisms, and correcting model outputs through validation against observations. Overcoming these challenges will position AI as a central pillar in advancing smart water quality governance, safeguarding water security, and achieving sustainable development goals.



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1. Introduction

The United Nations Sustainable Development Goal 6 (SDG 6) aims to ensure access to water and sanitation for all and promote their sustainable management. This goal emphasizes not only the availability of water resources but also the critical importance of water quality [1]. Amidst intensifying global urbanization, industrialization, and climate change, water pollution has emerged as an increasingly severe problem, placing immense pressure on water quality management [2,3]. Traditional water quality monitoring methods, which primarily rely on manual sampling and periodic laboratory analysis, suffer from several limitations. They are often inefficient, restricted in spatial coverage, and incapable of providing real-time responses to sudden pollution events. Consequently, these conventional approaches fail to meet the demands of modern water environment management for high-frequency, dynamic monitoring and rapid response capabilities [4]. Thus, the deep

integration of Artificial Intelligence (AI) with the Internet of Things (IoT), Remote Sensing (RS), and big data analytics is increasingly recognized as a pivotal technological solution for enhancing the effectiveness of water quality monitoring and management [5].

AI is fundamentally reshaping the paradigm of water environment monitoring and management through its unparalleled capabilities [6]. It is projected that from 2024 to 2032, the size of the global water management market for AI will grow from 7.54 billion US dollars to 53.85 billion US dollars, at an annual growth rate of 27.85% [7]. Its primary advantage lies in the efficient processing of complex, high-dimensional water environment data. This enables near-real-time dynamic analysis, accurate identification of anomalous pollution events (e.g., sudden leaks or algal blooms), and forward-looking predictions of water quality trends based on historical data and environmental factors [8]. A detailed comparison of performance indicators for AI-driven and conventional water monitoring approaches is illustrated in Table 1. Ponnuru et al. [9] assessed the performance of four Machine Learning (ML) algorithms—Support Vector Machines (SVM), Decision Trees (DT), Artificial Neural Networks (ANN), and Random Forests (RF)—in water quality monitoring using IoT sensor data, RS imagery, and publicly available environmental datasets. The results showed that ANN achieved the highest accuracy (95.2%), followed by RF at 92.8%, SVM at 89.5%, and DT at 87.3%. The AI models effectively reduced the error rate by 30%, improved real-time monitoring efficiency by 40%, and significantly enhanced pollution detection capabilities, demonstrating the outstanding performance of AI in real-time pollution detection and data processing. However, a critical obstacle arises from the inherent “black-box” nature of many AI models, whose decision-making processes often lack transparency and intuitive explainability [10]. This lack of transparency poses a challenge when crucial decisions concerning water quality safety are required (such as identifying pollution sources, assessing health risks, or formulating remediation strategies), as decision-makers may hesitate to fully trust models with opaque reasoning processes. Furthermore, AI models, particularly Deep Learning (DL), are heavily dependent on large-scale, high-quality, and accurately labeled training datasets [5]. This significant data reliance implies that the predictive accuracy and stability of these models can be substantially compromised in situations lacking sufficient data or when encountering non-standard, unforeseen environmental conditions (like the emergence of rare pollutants or extreme weather events). Such limitations may result in misleading outcomes and pose risks to water environmental safety. Therefore, overcoming these bottlenecks—namely explainability and data dependency—is crucial for advancing AI from theoretical research to large-scale, high-reliability applications in the water quality domain.

Table 1. Comparison of Performance Indicators between AI-Based and Traditional Water Quality Monitoring Methods.

Performance Indicators	AI Methods	Traditional Methods	Ref.
Timeliness	Enables real-time data acquisition and processing	Results delayed due to manual sampling and lab analysis	[10,11]
Accuracy	High precision through automated calibration	Susceptible to human error and environmental interference	[12,13]
Predictive Capability	Accurately predicts trends using DL models	Relies on simple models with limited forecasting capability	[11,14,15]

Table 1. *Cont.*

Performance Indicators	AI Methods	Traditional Methods	Ref.
Anomaly Detection Capability	Detects sudden changes in real time via pattern recognition algorithms	Infrequent sampling delays anomaly identification	[16,17]
Decision Support	Provides real-time, data-driven recommendations	Depends on periodic reports and expert judgment	[18–20]
Data Integration	Automatically integrates data from multiple sources	Manual integration leads to fragmented data	[14,21]
Cost	Reduces long-term operational costs via automation	High recurring costs in labor, analysis, and maintenance	[22,23]

Against this backdrop, this review systematically analyzes the current applications and recent advancements of AI in water quality monitoring, prediction, and management. The focus is placed on key technological areas such as real-time monitoring, predictive modeling, and management optimization. Specifically, it explores how AI enhances real-time monitoring accuracy, expands spatiotemporal coverage, improves prediction precision, and supports intelligent decision-making in water quality management. This article does not comprehensively cover all AI applications in water quality research, but instead focuses on current significant developments, highlighting successful experiences and limitations, and providing reference pathways for future research.

2. AI for Water Quality Monitoring

2.1. Application of AI in Real-Time Water Quality Monitoring

Traditional water quality monitoring methods, which rely on manual sampling and laboratory analysis, are inherently limited by low timeliness and insufficient spatiotemporal resolution [12]. With recent technological advancements, IoT-based sensors now facilitate continuous monitoring of various physical, chemical, and biological water quality parameters, transmitting data in real time to support more responsive and data-driven water management [17]. However, sensor data remains vulnerable to environmental disturbances, fluctuating hydrological conditions, and technical issues such as drift, malfunction, and data loss, particularly in complex environments (e.g., underwater or outdoor settings), which poses challenges to the accuracy and reliability of monitoring results [24,25]. Integrating AI—particularly ML algorithms—offers a promising solution to these challenges [26,27]. AI algorithms can intelligently process raw sensor data by performing real-time calibration, error correction, anomaly detection, and denoising, thereby significantly improving data quality and monitoring accuracy [8,28]. Additionally, AI's powerful pattern recognition capabilities enable automatic diagnosis of sensor faults, self-calibration, and data compensation, thereby enhancing the stability and robustness of monitoring systems [29]. The deployment of AI models at sensor terminals (i.e., edge computing) allows for localized intelligent analysis of data, reducing transmission delays, improving response speed, and granting sensors the ability to adapt to environmental changes and optimize autonomously. This collaborative integration of AI, sensors, and IoT not only greatly improves the accuracy, real-time performance, and intelligence of water quality monitoring but also effectively compensates for the inherent limitations of sensors under complex conditions [16,30]. Table 2 summarizes current applications of AI algorithms in water quality monitoring,

from traditional ML algorithms such as SVM and K-Means to DL models like Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM). AI technology is widely applied in key tasks such as water quality classification, pollutant concentration prediction, image-based pollution detection, and equipment fault diagnosis. These systems typically integrate IoT platforms with various sensors and combine RS or image data to enable low-cost, real-time remote data collection. Their significant advantages include high accuracy, real-time monitoring, low deployment costs, and strong environmental adaptability. The application fields cover various water environments such as drinking water, wastewater treatment, lakes, and aquaculture, significantly promoting the intelligent development of water quality monitoring and laying the foundation for building efficient water environment monitoring systems. Nonetheless, model interpretability and system-level integration remain critical challenges warranting further research.

Table 2. Applications of AI Methods in Water Quality Monitoring. (Abbreviations: DO = Dissolved Oxygen; TDS = Total Dissolved Solids; ORP = Oxidation-Reduction Potential; EC = Electrical Conductivity; COD = Chemical Oxygen Demand; BOD₅ = 5-day Biochemical Oxygen Demand; TN = Total Nitrogen; TP = Total Phosphorus; NH₄⁺ = Ammonium nitrogen; TSS = Total Suspended Solids; WQI = Water Quality Index.).

AI Methods	Task	Data Source	Parameters	Results	Ref.
SVM, ANN	Classification; Source prediction	IoT-based sensor collection system	Temperature, pH, Turbidity, DO, TDS, ORP, EC	Real-time monitoring; Remote transmission; Self-feedback and self-regulation; Low cost; Portable	[31]
K-Means	Classification; Automatic clustering analysis of sensor data	Underwater robot (sensors)	pH, Temperature, Turbidity, EC	Real-time underwater monitoring; Remote control & data upload; Suitable for large water bodies; Sustainable deployment; Low cost, minimal maintenance, compact design	[32]
RF, SVM, NN	Classification	Multispectral sensors	<i>Escherichia coli</i> (<i>E. coli</i>)	High accuracy	[16]
PSO-SVR	WQI prediction	Hyperspectral RS, In-situ data	Comprehensive water quality parameters (pH, DO, COD, BOD ₅ , TN, TP, TDS, etc.)	High-precision; Suitable for large-scale dry area water quality estimation	[33]
LSTM	Real-time monitoring (aquaculture)	Buoy sensors	Temperature, Salinity, DO, Flow velocity	Low cost; Real-time data; Short-term prediction	[22]
CNN	Real-time monitoring	Smartphone camera-captured sensor images	<i>E. coli</i>	High accuracy (99.99%); Objective; Simple operation	[34]
CNN	Water image classification (pollution)	IoT-collected water images	Pollution type visual features (e.g., texture, oil stains, animal carcasses, etc.)	High precision; Enhanced features; Real-time monitoring/feedback	[35]
Boosting-IPW-PLS	Wastewater monitoring	Sensors	COD, TSS, Oil and Grease	High precision; Handles noise; Online monitoring	[36]

Table 2. Cont.

AI Methods	Task	Data Source	Parameters	Results	Ref.
ANN	Sensor fault detection and replacement control	Sensors	NH ₄ , pH, ORP, DO, TSS	High accuracy; Rapid fault ID; Automatic control replacement after fault detection	[37]
ANN	Prediction and disinfectant optimization control	Laboratory monitoring + sensors	<i>E. coli</i>	High-precision; Optimizes peracetic acid disinfectant usage	[38]

2.2. AI Combined with RS and Unmanned Monitoring: Expanding New Dimensions in Water Quality Monitoring

While integrating AI with in-situ sensors has improved real-time monitoring accuracy, traditional monitoring stations still face limitations in spatial coverage and deployment flexibility. The adoption of RS technology and UMP offers broader spatial perspectives and greater operational flexibility in water quality monitoring.

RS technology is particularly valuable for providing large-scale, near-synchronous observations of surface water bodies, which is crucial for assessing the overall condition of vast water areas [39]. AI algorithms enable efficient analysis of RS imagery, allowing for precise extraction of surface water information, and enabling rapid identification of water quality issues such as algal blooms and pollution [40]. Chebud et al. [41] demonstrated that using ANN to process Landsat TM RS data allows for the monitoring of parameters such as chlorophyll-a (Chl-a), turbidity, and TP, providing a low-cost and efficient method for large-scale water body ecological assessments.

Unlike RS, which provides a macro perspective, the core advantage of UMP lies in their spatial deployment flexibility and the ability to perform close-range detection in specific areas. Unmanned systems, including Unmanned Aerial Systems (UAS), Unmanned Surface Vehicles (USV), and Unmanned Underwater Vehicles (UUV), significantly extend the spatial flexibility and accessibility of monitoring. These systems can access areas that are difficult to reach by traditional methods, such as remote water bodies, hazardous pollution zones, or disaster-affected areas [42]. Vasudevan and Baskaran [43] proposed an IoT-based USV system for real-time monitoring of parameters such as pH, temperature, and dissolved oxygen (DO) in remote or hard-to-reach water bodies. This system demonstrated advantages in low cost, low energy consumption, and high coverage, effectively overcoming the limitations of manual sampling and fixed monitoring stations in terms of coverage and maintenance. Ryu [42] developed an unmanned aerial platform that integrates monitoring, sampling, and visualization functions. Using the commercial drone DJI Matrice 600 Pro, equipped with 3D-printed sampling devices and sensors, this platform can monitor temperature, pH, conductivity, and DO in real time in remote or hazardous water areas. It also collects 1 L of water samples and transmits data to the cloud for visualization analysis via IoT technology. This system demonstrated the advantages of low cost, high mobility, and wide coverage, offering innovative support for environmental management.

In particular, UAS equipped with sensors for RS operations (i.e., drone RS) offers unique advantages for water quality monitoring [44]. Compared to satellite RS, drone-based RS can quickly obtain high spatial resolution image data at a lower cost and with greater flexibility, making it particularly suitable for detailed monitoring and investigation of small water bodies, shoreline areas, or specific pollution events. It effectively captures local water quality changes and supports high-frequency repeated observations.

Despite these technological advancements, current applications remain constrained by several key limitations: Most existing studies primarily focus on optically active variables,

including Chl-a, colored dissolved organic matter (CDOM), TSS, and turbidity. However, many important water quality variables, such as pH, total nitrogen (TN), ammonia nitrogen ($\text{NH}_3\text{-N}$), nitrate nitrogen ($\text{NO}_3\text{-N}$), and dissolved phosphorus (DP), have not been well studied due to their weak optical properties and low signal-to-noise ratio. Additionally, data processing complexity: The massive data generated by RS imagery and unmanned devices places high demands on the computational capacity of AI algorithms. For example, satellite multispectral imagery often contains noise (e.g., cloud interference), necessitating complex preprocessing procedures. Equipment costs and maintenance: The procurement and maintenance costs of drones, USVs, and UUVs are relatively high [45], and they are easily affected by environmental factors (such as strong winds and water currents). For instance, the stability of USVs in harsh weather conditions still requires improvement. Spatial and depth limitations: Satellite RS has difficulty penetrating the water surface and is limited to monitoring the surface layer; although UUV can dive underwater, their coverage is limited, making large-scale deep monitoring challenging. Lack of technical standardization: The data formats from different RS platforms and unmanned devices are not unified, meaning that AI models need to be customized for specific data sources, which limits the universality of the systems.

2.3. Multi-Source Data Fusion and Intelligent Analysis

Although advanced technologies such as RS and UMP have greatly enhanced the frequency of water quality data collection and spatial coverage, relying on a single data source or simple data overlay is far from sufficient for transforming this abundant data into accurate assessments and effective decision-making. Given the complex and ever-changing water environment and the vast, heterogeneous data it generates, the combination of multi-source data fusion and intelligent analysis has become a key solution to overcoming bottlenecks and improving the intelligence level of water quality monitoring systems [46,47].

In the field of water quality monitoring, multi-source data fusion refers to the systematic integration of information from different channels to obtain a more comprehensive, reliable, and accurate understanding of water quality conditions than any single source alone [48,49]. These data sources can include:

High temporal resolution in-situ sensor data: Real-time or near-real-time measurements from fixed monitoring stations or mobile platforms (such as USV), including parameters like pH, DO, temperature, turbidity, etc.

Wide-area RS imagery data: Data from satellites or drones that provide spatial distribution information for surface water parameters such as Chl-a, suspended solids, water color index, and surface temperature.

High-precision laboratory analysis data: Chemical and biological tests conducted on collected water samples, such as for heavy metals, specific organic pollutants, and nutrients.

Other data: Including meteorological data (precipitation, temperature), hydrological data (flow rate, velocity), geographic information (land use, point-source pollution distribution), and socio-economic data.

By integrating these complementary data sources, the limitations of single data sources can be effectively overcome [50]. For example, RS can quickly locate potential problem areas, after which unmanned platforms can be deployed for close-range verification and sampling. This data can then be combined with continuous monitoring data from fixed stations and precise analysis results from laboratories, forming a spatially and temporally continuous, integrated monitoring network that combines both macro and micro perspectives [51].

The fused multi-source data typically exhibit characteristics of big data, including high volume (Volume), a variety of data types (Variety), fast velocity (Velocity), and low-value density (Value), making it challenging for traditional data processing methods to handle

effectively [52]. In such cases, intelligent analysis technologies, represented by AI and ML, become particularly important. AI algorithms, especially DL models, excel at processing high-dimensional, nonlinear complex data and are capable of:

Automatic identification and extraction of hidden patterns: Discovering complex correlations and trends in the fused data that are difficult to detect through traditional methods.

Filling in data gaps: Estimating missing spatiotemporal data points by leveraging the relationships between available data.

Improving prediction accuracy: Building more robust water quality prediction models that provide early warnings for potential risks.

Supporting intelligent decision-making: Offering more reliable data support and decision recommendations for water environment management.

Jiang et al. [14] provide an excellent example of this approach. They successfully integrated data from various sources, including environmental, social, water quantity, and water quality data, using DL models to predict key pollution indicators in urban sewage networks (BOD₅, COD, NH₄⁺-N, TN, TP). The results showed that the model's coefficient of determination (R^2) ranged from 0.85 to 0.91, significantly outperforming traditional statistical methods. This clearly demonstrates the immense potential of AI-driven multi-source data fusion in extracting data value and improving water quality prediction accuracy.

Therefore, developing and applying multi-source data fusion technologies, combined with powerful AI-driven intelligent analysis capabilities, is a critical step in advancing modern water quality monitoring from “data collection” to “intelligent insights”. This not only greatly enhances the accuracy and reliability of monitoring results but also provides unprecedented decision support for water resource protection and environmental management.

3. The Application of AI in Water Quality Prediction

3.1. Water Quality Prediction Based on Historical and Real-Time Data

The development of water quality prediction models has evolved from traditional statistical regression to classic ML algorithms, and further to DL models. ML models, such as SVM and RF, have demonstrated a superior ability to predict key indicators like DO, chemical oxygen demand (COD), and TN compared to traditional statistical regressions (e.g., multiple linear regression (MLR), Autoregressive Integrated Moving Average ARIMA), particularly when dealing with complex nonlinear relationships between water quality parameters [53]. With advancements in time series data processing capabilities, water quality prediction models have gradually become more refined. He et al. [19] categorized these models into three patterns: the point-to-point (P2P) model, which uses synchronous data to predict the current state; the sequence-to-point (S2P) model, which uses historical sequences to predict future single points; and the sequence-to-sequence (S2S) model, which predicts future sequences based on historical data. The article discusses traditional ML methods (such as SVM, RF, multilayer perceptron neural networks (MLPNN), etc.) as well as DL methods (such as LSTM, CNN, etc.). The research indicates that DL models, especially LSTM and Gated Recurrent Units (GRU), have been widely applied in water quality prediction in recent years due to their advantages in handling time-series data. LSTM, an improved version of Recurrent Neural Networks (RNN), can effectively address the long-term dependency problem. In water quality prediction, LSTM can predict future water quality changes by inputting historical water quality data, particularly performing well with time-series water quality data [19,54]. AI technologies have also advanced the timeliness of water quality information acquisition. While traditional real-time monitoring relies on physical sensors, AI models can construct real-time simulation and prediction

systems, providing rapid assessments of current and future water quality conditions. One example is Bang et al. [55], who developed a real-time water quality simulation toolbox using ML and application programming interfaces (APIs) to simulate and predict parameters such as DO, TN, and TP, presented in a graphical user interface (GUI). The near-real-time simulation and prediction results provided by this tool are crucial for timely responses to water quality changes and pollution event early warnings.

In addition to conventional water quality parameters, ML has gradually been applied to the study of pollutants that are difficult to measure online, such as emerging contaminants [56,57]. This includes the identification, environmental behavior prediction, toxicity analysis, and removal technology optimization of emerging pollutants like per-fluoroalkyl substances, nanoparticles, antibiotic resistance genes, and microplastics [56]. By processing complex, high-dimensional, and unstructured data, ML enhances the accuracy and efficiency of pollutant detection, offering new approaches and methods for environmental pollution management.

To further improve prediction accuracy and address the complexity of real-world water systems, researchers have explored hybrid model strategies [58,59]. Rajaei et al. [60] reviewed the application of both single and hybrid AI-based models in river water quality prediction. The article evaluated several AI models, including ANN, Genetic Programming (GP), Fuzzy Logic (FL), and SVM, with particular emphasis on hybrid models such as Wavelet-Neural Networks (WANN) and Wavelet-Support Vector Regression (WSVR) for their advantages in enhancing prediction accuracy. The study found that ANN and WANN were the most commonly used models, especially excelling in predicting DO and suspended sediment load (SSL). Compared to traditional regression models (e.g., MLR, ARIMA), AI models are better at handling complex nonlinear relationships and noisy data.

Despite the powerful data mining capabilities demonstrated by AI in water quality prediction, its application still faces significant challenges. Purely data-driven models are highly sensitive to the quality and quantity of data, and they are constrained by issues such as missing data, outliers, and uneven spatial and temporal coverage [15]. The models themselves also exhibit uncertainties, such as the subjectivity involved in selecting indicators and assigning weights when constructing a comprehensive WQI, which may lead to predictive bias [21]. More critically, many existing AI models tend to focus on spatial points or time series predictions, making it difficult to ensure spatiotemporal consistency in the underlying physical processes. Furthermore, the “black-box” nature of the models limits their generalization ability and mechanistic interpretability, especially when confronted with long-term macro factors like climate change or sparse data scenarios, where their reliability is put to the test. On the other hand, traditional process-driven models (e.g., hydrodynamic-water quality models) are based on physical, chemical, and biological principles, and can simulate the complex two-dimensional or three-dimensional spatiotemporal dynamics of water bodies. These models have a solid mechanistic foundation and good interpretability. However, they tend to be structurally complex, computationally expensive, and require stringent parameterization and boundary condition settings, with the calibration process potentially introducing uncertainties. Therefore, coupling data-driven AI models with process-based physical models is an extremely promising research direction.

3.2. AI Coupled with Process-Based Models (PBM)

Traditional water quality models are primarily based on mathematical descriptions of physical, chemical, and biological processes. While they offer good mechanistic interpretability, they often face bottlenecks such as low computational efficiency and difficulty in parameter calibration when simulating complex, nonlinear water environment systems [61]. On the other hand, purely data-driven AI models, while possessing powerful fitting ca-

pabilities, lack physical constraints, potentially leading to predictions that do not align with natural laws [62]. The coupling of AI with process models combines the strengths of both approaches, enabling more efficient and accurate water quality prediction and management [63]. The core idea behind this “hybrid modeling” or “Process-Guided AI” (PGAI) is to leverage AI’s powerful data learning capabilities to capture complex patterns while incorporating the physical mechanisms of process models as constraints or prior knowledge. This approach not only significantly enhances prediction accuracy, robustness, and generalization ability, ensuring the physical consistency of results, but also effectively compensates for the shortcomings of purely data-driven models—such as the lack of mechanistic grounding—and the challenges associated with calibrating pure process models, which can be computationally expensive. It is particularly valuable in scenarios with sparse data [64].

As shown in Figure 1, effective approaches for coupling AI with process models include parameter optimization, hybrid modeling, and process-guided deep learning (PGDL) [65].

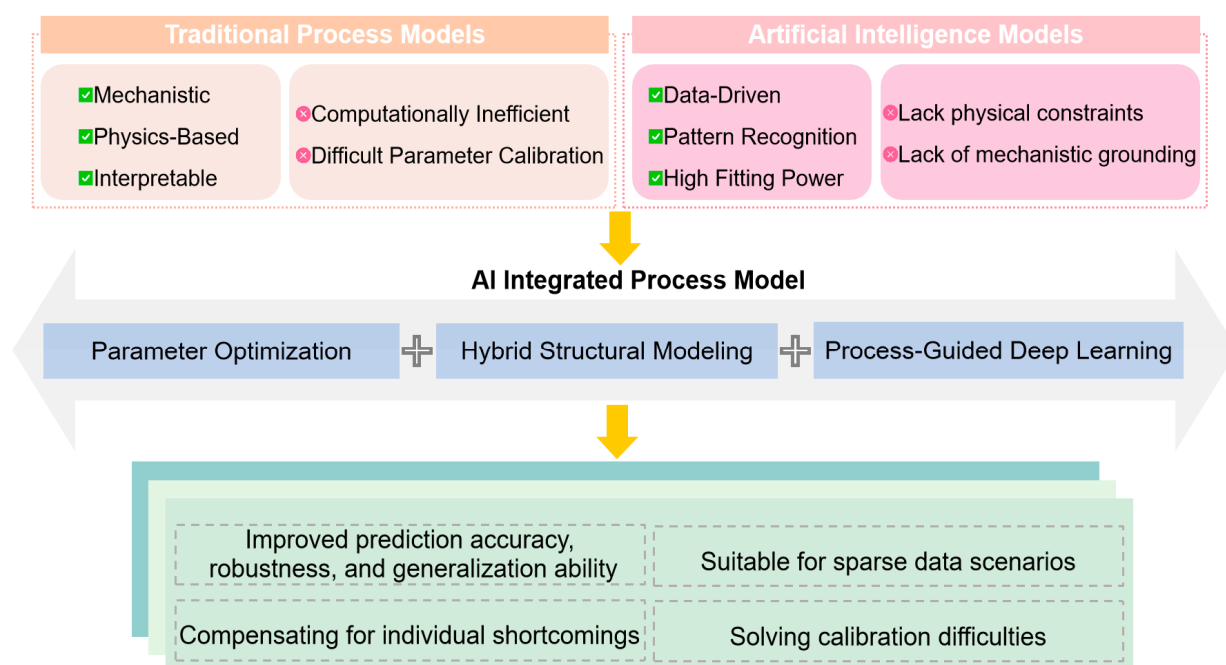


Figure 1. Integration of AI and PBM for Enhanced Water Quality Prediction.

Parameter Optimization: AI optimization algorithms (e.g., genetic algorithms, particle swarm optimization) are used to automate and efficiently calibrate key parameters of traditional process models, replacing the tedious manual parameter adjustment process.

Hybrid Structural Modeling: Process models are combined with AI models (such as SVM) through serial, parallel, or feedback loops. The process model provides the physical framework or initial predictions, while the AI model corrects biases or handles complex nonlinear components.

PGDL: Physical process equations (e.g., differential equations) are directly embedded into the structure or loss function of DL models, allowing the model to simultaneously learn data patterns while adhering to physical laws during the training process.

This coupling strategy greatly enhances the ability to simulate the complex processes of water environments. For example, by combining DL with physically coupled hydrodynamic models, it becomes possible to more accurately characterize the spatiotemporal migration and transformation paths of pollutants and their concentration dynamics in

complex river networks, significantly improving prediction accuracy and stability (as summarized in Figure 1) [66]. Similarly, integrating process models with data-driven models has been shown to effectively improve the prediction accuracy of emerging pollutants and allow for rapid assessments based on easily measurable water quality indicators, reducing computational costs [67]. Wang et al. [68] proposed a new framework that combines a process-based water quality model with a Modified Local Model (MLM) based on chaos theory for data assimilation, aimed at enhancing real-time forecasting of salinity and DO in Singapore's coastal waters. Their approach significantly reduced the root-mean-square error (RMSE) in daily forecasts over 1–5 days, achieving RMSE values as low as 0.23 for DO and 0.36 for salinity. This demonstrates that the method effectively improves forecast accuracy for both short-term and long-term predictions. Further, Wang et al. [69] developed an integrated modeling framework at the watershed scale, combining hydrological, hydraulic, and water quality models, and improved water quality predictions using data assimilation techniques. In its application to the coastal waters of Singapore, this method significantly enhanced the prediction accuracy of Chl-a. Specifically, for a 2-day forecast, the RMSE of Chl-a decreased from 5.96 to 2.49, a reduction of 58%; for a 2-hour forecast, the RMSE decreased by 48%. These results indicate that data assimilation techniques substantially improved the model's predictive capability, especially in real-time forecasting. This approach is not only applicable to long-term water quality planning and management but also to real-time water quality change predictions and event-driven water ecosystem changes. Additionally, the method was successfully applied to both coastal and freshwater systems in Singapore, providing a powerful tool for water quality management. A notable example of a successful AI-coupled PBM is the hybrid approach developed by Tong et al. [67], which combined a PBM with data-driven models (DDMs) to predict emerging contaminants (ECs) such as Bisphenol A (BPA) and N,N-diethyltoluamide (DEET) in a tropical reservoir. Their study developed 36 DDMs using ANN and RF, achieving high predictive accuracy. The relatively high R^2 , Nash-Sutcliffe Efficiency (NSE), low RMSE, and average relative deviation (ARD (%))—with $R^2 \sim 0.84$, $NSE \sim 0.91$, $RMSE \sim 0.12$, $ARD (\%) \sim 2.00$ —indicate that the models are highly capable of accurately predicting EC concentrations in the reservoir. Notably, this hybrid model reduced computational time by over 99% compared to standalone PBM, completing one-year simulations in approximately one minute, as opposed to 8 hours for PBM alone. This efficiency, coupled with the use of easily measurable water quality indicators (e.g., TN, TP, Chl-a), demonstrates the potential of hybrid models to provide rapid, cost-effective predictions for hard-to-measure pollutants, thereby supporting real-time water quality management in urban reservoirs. Likewise, coupling climate change models with AI models enables more reliable evaluations of the long-term potential impacts of future climate scenarios on water quality, providing scientific support for water resource adaptive management [70].

Despite the promising prospects, the widespread application of AI-coupled process models still faces challenges. The effective integration of heterogeneous data is a primary issue, as the two types of models often require different data types, resolutions, and formats. Moreover, the choice and optimization of coupling methods lack a systematic evaluation framework, requiring clear identification of the applicable boundaries and performance strengths and weaknesses of different approaches. Additionally, the computational burden imposed by complex coupled models is a major constraint.

Future research should focus on developing advanced data fusion technologies, establishing standardized coupling frameworks and evaluation systems, and using model simplification and high-performance computing (e.g., parallel computing, cloud computing) to improve computational efficiency. Through continuous technological innovation and interdisciplinary collaboration, AI-coupled process models are expected to become a

more precise, efficient, and reliable core technological support in the fields of water quality prediction and smart water management, contributing key insights to ensure the health of water environments and the sustainable use of water resources.

4. The Application of AI in Smart Water Environment Management

AI is profoundly transforming water quality management, driving the shift from traditional passive responses to proactive prevention, precise prediction, and optimized decision-making. The integration of AI technology has significantly enhanced monitoring capabilities, operational efficiency, pollution control levels, and the scientific basis of overall management decisions [6].

The combination of AI and the IoT is a key pathway to achieving dynamic and efficient water management [17]. By integrating various water quality parameters collected in real-time by IoT sensors deployed in water bodies, AI algorithms—particularly ML and DL—are able to efficiently process these large-scale, heterogeneous data sets. This enables dynamic monitoring to track water quality changes in real time, trend forecasting based on existing data and historical patterns to predict future water quality conditions, and early warning systems to identify potential risks such as algal blooms or heavy metal contamination. These advancements strongly support the fundamental shift in management models from “reactive processing” to “proactive prevention”. At the same time, AI can optimize operations and maintenance by analyzing sensor data and equipment status, such as predicting potential failures (predictive maintenance) and optimizing maintenance schedules, thereby reducing equipment downtime and ensuring the stable operation of monitoring and treatment facilities [71]. Examples of successful integrated systems include the Korea Water Resources Corporation, which uses IoT, Geographic Information System (GIS), and AI for real-time monitoring, leak detection, and flood management, as well as the PUMAGUA project in Mexico, which leverages AI and IoT for water conservation and water quality improvement. These systems enable decision-makers to prioritize key areas or develop forward-looking management strategies based on real-time conditions and future forecasts [72,73].

AI has significantly enhanced the ability to identify and track pollution sources, a critical step in effective pollution management. By integrating and analyzing multi-source data from sensor networks, RS imagery, social media information, and official environmental reports, AI can more accurately pinpoint the origin of pollution events [51], simulate the migration and diffusion paths of pollutants in water bodies [63], and even quickly identify pollutant types by combining technologies like chemical spectroscopy [74]. The integration of AI with citizen science initiatives, such as crowdsourced data collection, offers a promising avenue for expanding spatial and temporal coverage of water quality observations [75]. Smartphone-enabled image recognition, low-cost sensors, and cloud platforms can empower local communities to participate in environmental monitoring, while AI algorithms can assist in validating, filtering, and analyzing these decentralized datasets to improve data quality and usability for decision-making [76]. This is particularly crucial for tracing the origins of complex ECs. Additionally, AI can help analyze the contributions of various pollution sources, such as industrial emissions and agricultural runoff, to the contamination of water bodies, providing a scientific basis for developing more targeted pollution control strategies [77].

The integration of AI with Decision Support Systems (DSS) has provided powerful intelligent support for water quality management. AI models, by learning from vast amounts of historical data and integrating real-time information, can not only predict water quality changes in real time but also offer data-driven optimization recommendations for water resource allocation and pollution control measures [20]. Furthermore, AI can assist in

evaluating the potential effects of different management decisions through simulations, aiding in the selection of the best approach. This intelligent support makes the water quality management decision-making process more scientific, efficient, and accurate, laying a solid foundation for ensuring water resource safety and sustainable use. To visually summarize the overall process, Figure 2 illustrates a conceptual framework of AI-driven water quality management. This framework integrates data acquisition from IoT sensors, RS platforms, and citizen science; performs data fusion and intelligent analysis using ML and DL algorithms; and supports decision-making processes including risk forecasting, pollution source identification, and infrastructure optimization. The system operates as a dynamic feedback loop, enabling proactive and adaptive responses in real-time water governance.

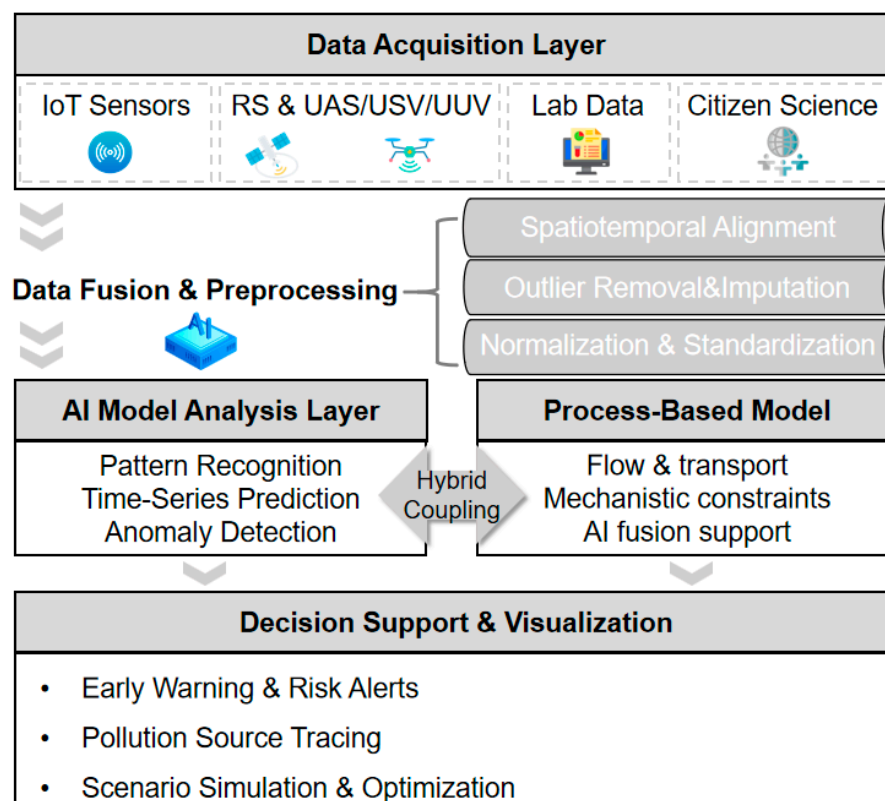


Figure 2. Conceptual framework of AI-driven water quality management.

5. Conclusions and Prospects

AI is rapidly emerging as a transformative force in modern water quality monitoring, prediction, and management, offering robust capabilities to address growing pressures on global water resources. As reviewed, the integration of AI with IoT, RS, and UMP has substantially advanced real-time water quality assessment, expanding spatiotemporal coverage and enabling comprehensive multi-source data analysis.

In predictive modeling, cutting-edge AI approaches, particularly DL and hybrid models combined with process-based approaches (i.e., PGAI), demonstrate superior performance in capturing complex nonlinear dynamics and predicting water quality trends compared to traditional methods. Moreover, AI is driving a paradigm shift in water quality governance by enabling a transition from reactive measures to proactive management. This includes dynamic early-warning systems, precise pollutant source identification, and optimized data-driven decision-making, significantly enhancing operational responsiveness and long-term strategic planning.

Despite these significant strides, realizing the full potential of AI in the water quality domain hinges on overcoming several persistent challenges (Figure 3). Critical obstacles include the scarcity of high-quality and standardized datasets essential for robust model training; the inherent lack of transparency in many complex AI models, which impedes trust and adoption for critical decisions; the ongoing difficulty in reliably monitoring hard-to-measure pollutants like emerging contaminants in real-time; and the complexities associated with integrating diverse data sources, platforms, and models seamlessly.

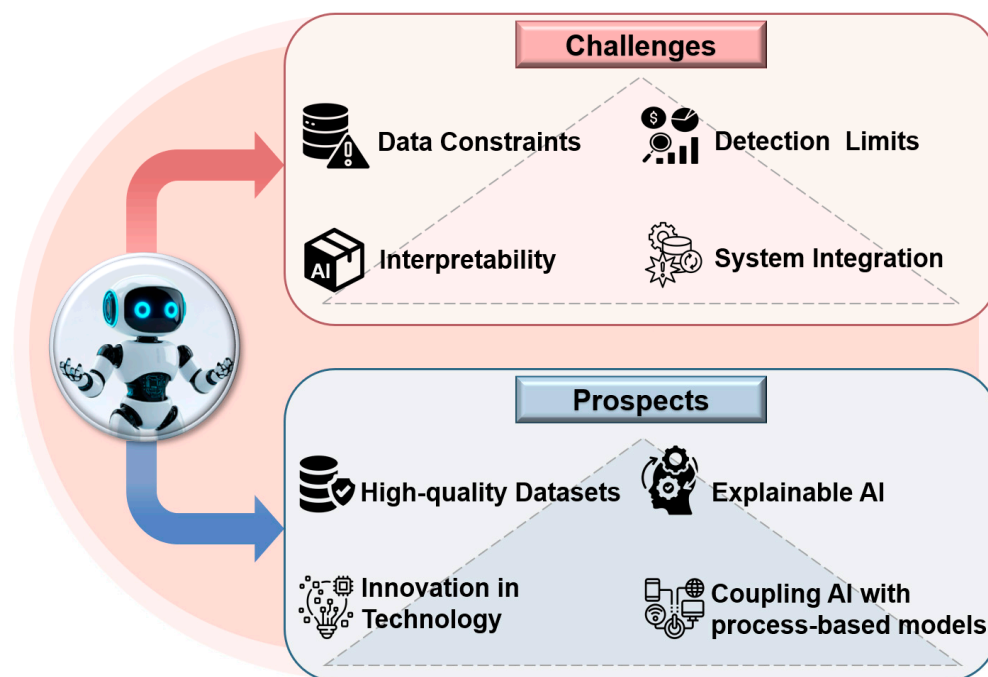


Figure 3. Overcoming Challenges: Prospects for Advancing AI in Water Quality Monitoring and Management.

Addressing these bottlenecks is paramount for future progress. Research and development priorities should include: (1) constructing high-quality, standardized open-access datasets to ensure robust model training and enable effective benchmarking across studies; (2) developing explainable AI (XAI) models to enhance transparency, build stakeholder trust, and facilitate adoption for critical water management decisions; (3) strengthening integration with digital twins and next-generation sensors to create dynamic virtual replicas and provide richer data streams for more comprehensive system understanding and real-time control; (4) improving the monitoring of trace and emerging pollutants, leveraging AI to enhance detection from complex sensor signals and predict the occurrence of these hard-to-measure substances; and (5) coupling AI with PBM to synergistically combine data-driven insights with mechanistic understanding, thereby improving predictive accuracy and physical consistency by optimizing input data, internal mechanisms, and correcting model outputs through validation against observations.

By tackling these challenges through sustained interdisciplinary innovation and collaboration, AI is poised to play an increasingly pivotal role. Successfully navigating these hurdles will unlock AI's capacity to significantly contribute to ensuring global water safety, achieving SDG 6, and ushering in an era of smarter, more precise, and adaptive water resource management.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
ANN	Artificial Neural Networks
APIs	application programming interfaces
ARD	average relative deviation
ARIMA	Autoregressive Integrated Moving Average
BOD ₅	5-day Biochemical Oxygen Demand
Boosting-IPW-PLS	Boosting-Iterative Predictor Weighting-Partial Least Squares
BPA	Bisphenol A
CDOM	Colored Dissolved Organic Matter
Chl-a	chlorophyll-a
CNN	Convolutional Neural Networks
COD	chemical oxygen demand
DDMs	data-driven models
DEET	N,N-diethyltoluamide
DL	Deep Learning
DO	dissolved oxygen
DP	dissolved phosphorus
DSS	Decision Support Systems
DT	Decision Trees
EC	Electrical Conductivity
ECs	emerging contaminants
FL	Fuzzy Logic
GIS	Geographic Information System
GP	Genetic Programming
GRU	Gated Recurrent Unit
GUI	graphical user interface
IoT	Internet of Things
K-Means	K-Means clustering
LSTM	Long Short-Term Memory
ML	Machine Learning
MLM	Modified Local Model
MLPNN	multilayer perceptron neural networks
MLR	multiple linear regression
NN	Neural Network
NH ₃ -N	Ammonia Nitrogen
NH ₄ ⁺	Ammonium nitrogen

NO ₃ -N	Nitrate Nitrogen
NSE	Nash-Sutcliffe Efficiency
ORP	Oxidation-Reduction Potential
P2P	point-to-point
PBM	process-based models (or Process-Based Models)
PGAI	Process-Guided AI
PGDL	process-guided deep learning
PSO-SVR	particle swarm optimization-support vector regression
R ²	coefficient of determination
RF	Random Forests
RNN	Recurrent Neural Networks
RMSE	root-mean-square error
RS	Remote Sensing
S2P	sequence-to-point
S2S	sequence-to-sequence
SDG 6	Sustainable Development Goal 6
SSL	suspended sediment load
SVM	Support Vector Machines
TDS	Total Dissolved Solids
TN	Total Nitrogen
TP	Total Phosphorus
TSS	Total Suspended Solids
UAS	Unmanned Aerial Systems
UMP	Unmanned Monitoring Platforms
USV	Unmanned Surface Vehicles
UUV	Unmanned Underwater Vehicles
WANN	Wavelet-Neural Networks
WQI	Water Quality Index
WSVR	Wavelet-Support Vector Regression
XAI	explainable AI

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